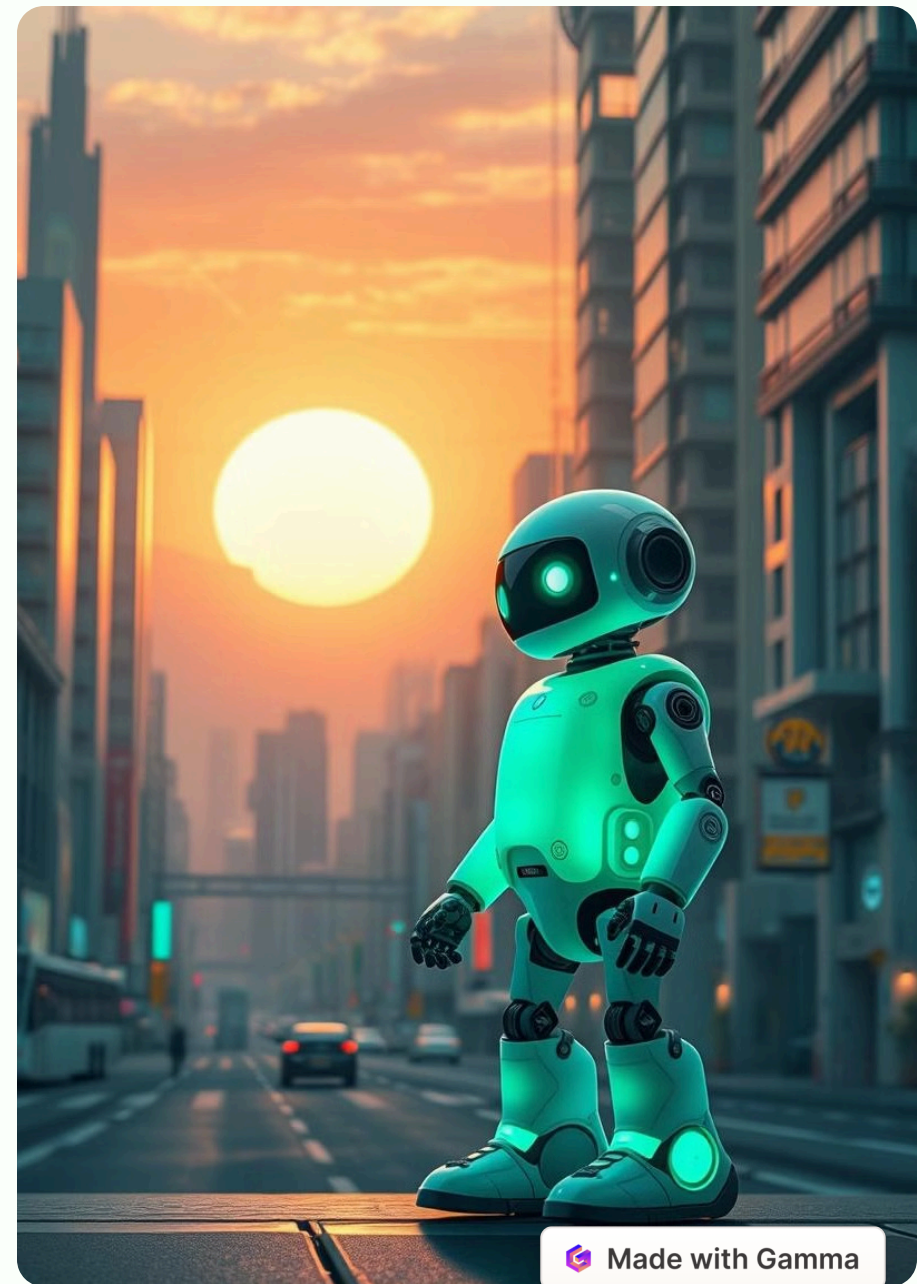
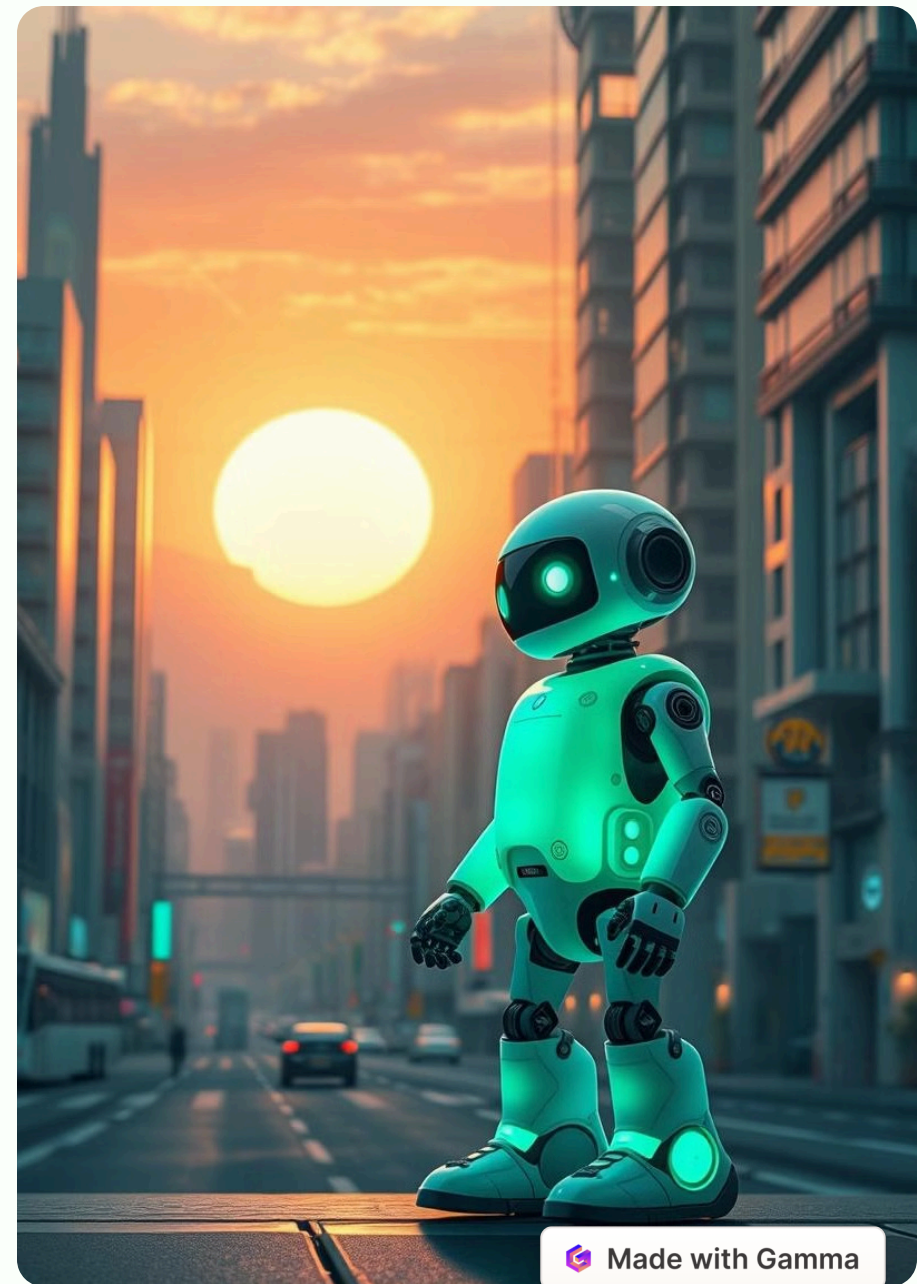


REINFORCEMENT LEARNING(RL)



Introduction to Reinforcement Learning

- RL is a branch of Machine learning
- RL focuses on agents
- take actions
- cumulative rewards



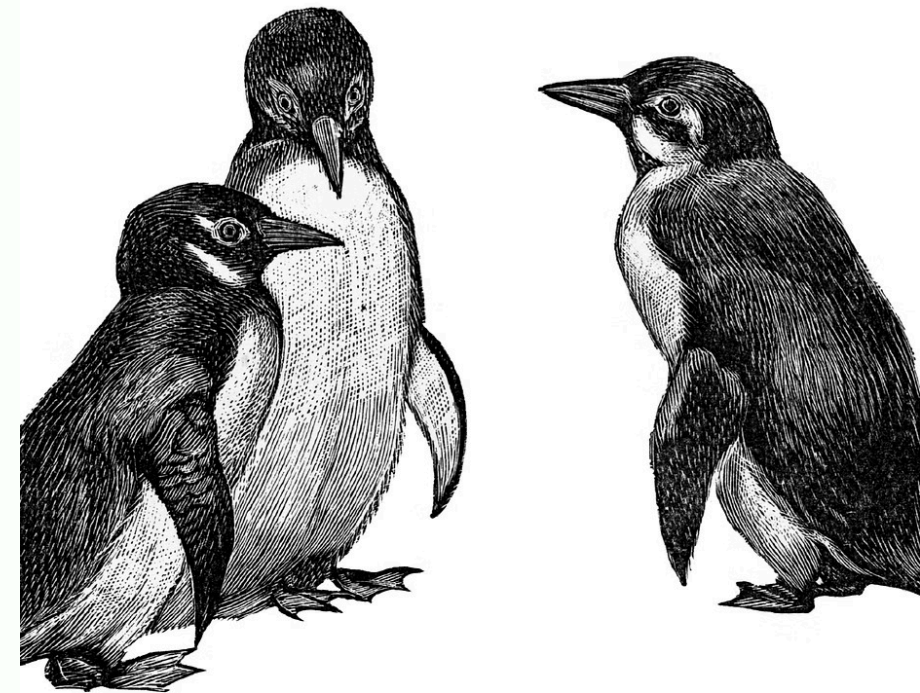
Reinforcement Learning(key concepts)

- Agent:
 - Assists in decision-making
 - take action
- Environment:
 - a simulation
 - interacts with agent
 - Optimize actions
- States:
 - current environment
 - agent
- Actions:
 - used by agents
 - change the state of an environment
- Rewards
 - provides feedback to agents
 - motivates to learn and explore
- Policy:
 - a function
 - maps environmental states to actions
 - defines agent behavior

REILLY®

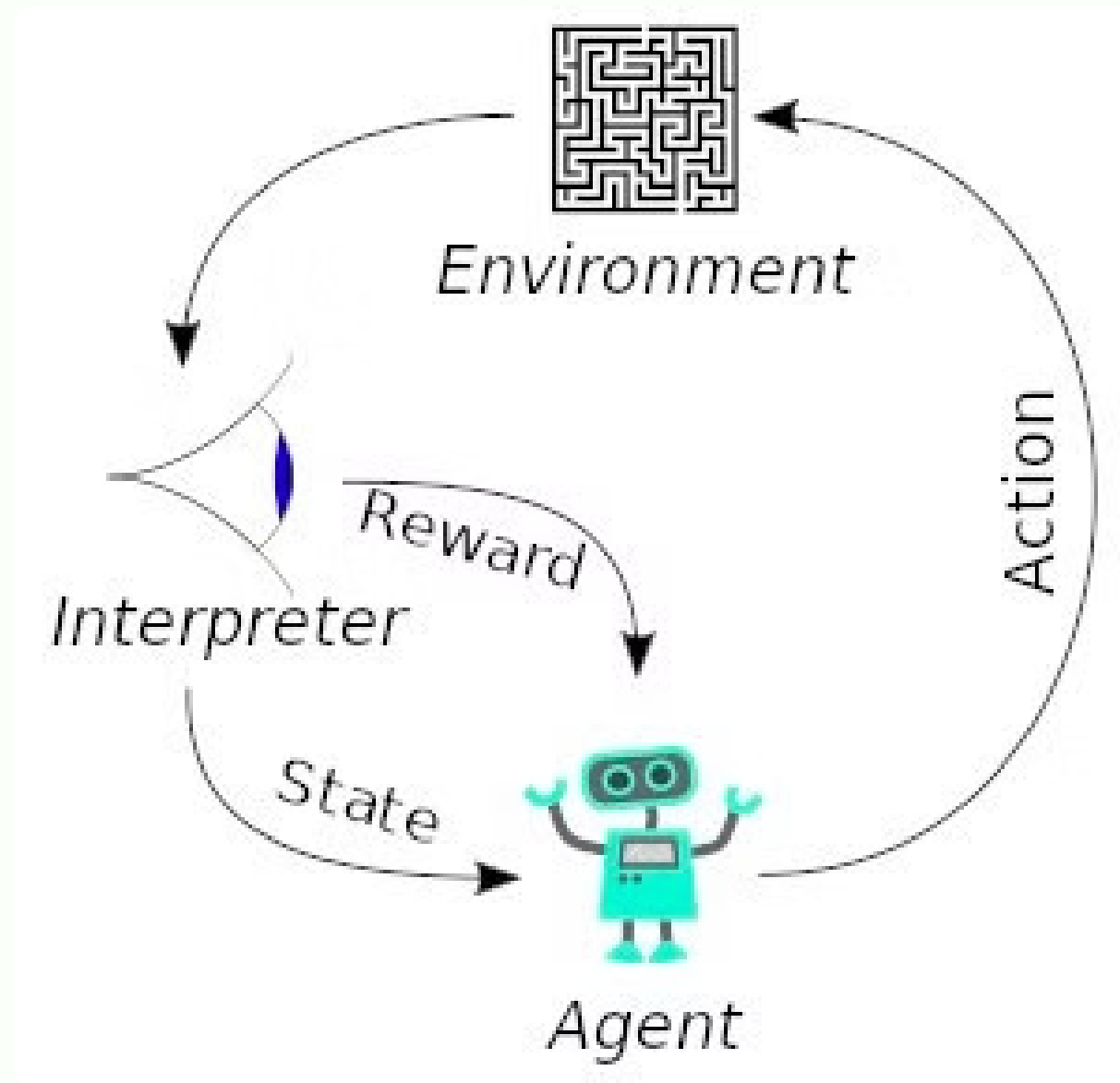
Reinforcement Learning

Industrial Applications of Intelligent Agents



Phil Winder, Ph.D.

How does reinforcement learning work?



REAL WORLD EXAMPLES

- Healthcare:
 - Chronic illnesses: cancer and HIV
 - analyses data collected from ICU
 - accurate and improved critical ongoing treatment
- robotics :
 - robotic arm
- self-driving cars:
 - automated parking policies
 - lane changing achieved via Q-learning

The Q-Value (Quality of Action)

Understanding Q-Value

- 2 points
- Take a specific action
- given state
- Q value or action-value is the expected value
- Q-value is used in Q-learning
- Q-learning = a RL algorithm
- Best action for each state

Storing Q-values

- 2D table
- rows = states
- columns = actions
 - Best action according to state.
→ This leads to optimizing rewards.

Exploration vs. Exploitation

Exploration

The agent tries new actions to discover potentially better options and expand its knowledge of the environment.

Exploitation

The agent leverages its current knowledge to select actions that are expected to yield the highest immediate reward.

Balancing Act

Striking the right balance between exploration and exploitation is crucial for the agent to learn effectively and make optimal decisions.

Q-Learning Algorithm

1

Initialization

The agent starts with initial Q-values, typically set to a default value or based on prior knowledge.

2

Observe State

The agent observes the current state of the environment and the available actions.

3

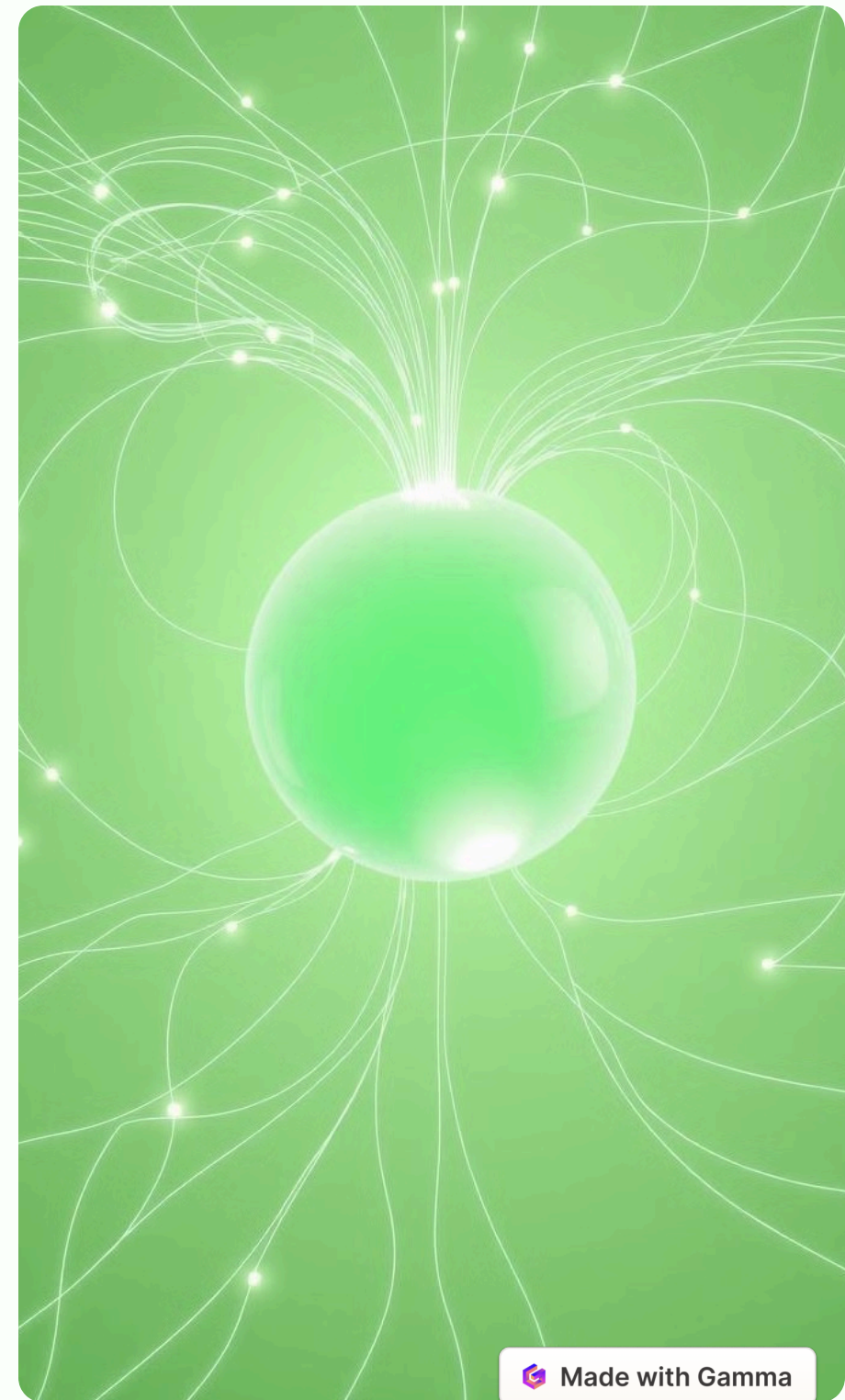
Select Action

The agent selects an action based on a balance between exploration and exploitation.

4

Update Q-Value

The agent updates the Q-value for the selected action based on the immediate reward and the expected future reward.



Formula for Q-learning algorithm

$$Q(s, a) = Q(s, a) + \alpha \left(r + \gamma \cdot \max_{a'} Q(s', a') - Q(s, a) \right)$$

where:

- s : Current state
- a : Current action
- r : Reward for taking action a in state s
- s' : Next state
- α : Learning rate
- γ : Discount factor (controls how future rewards are valued compared to immediate rewards)

Challenges

1 Sparse Rewards

Dealing with environments where rewards are infrequent or delayed, requiring more sophisticated exploration strategies.

3 Exploration-Exploitation Dilemma

Balancing the need to explore new options while also leveraging current knowledge to maximize rewards.

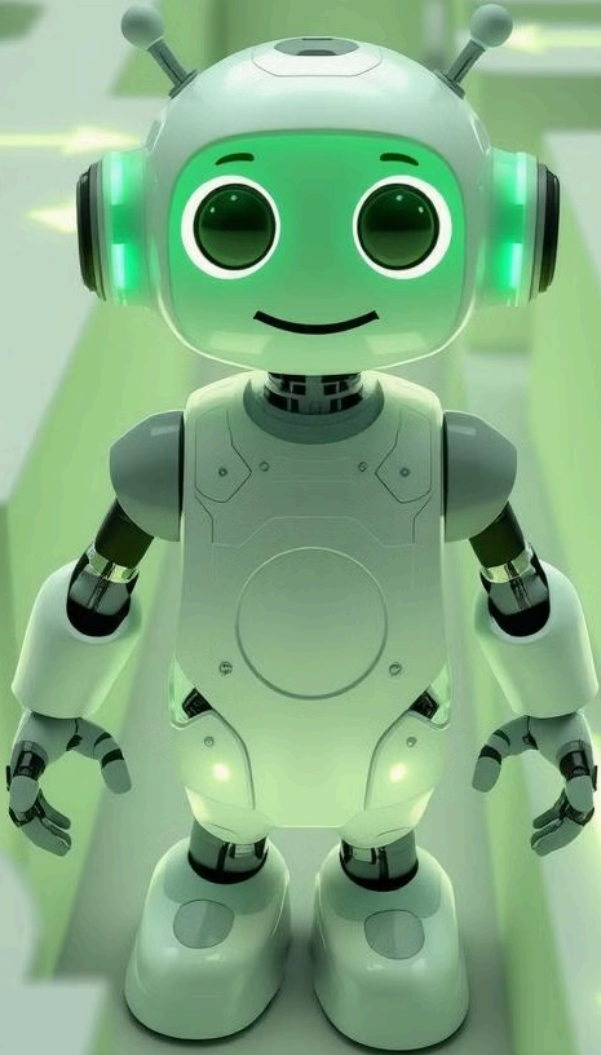
2 High Dimensionality

Handling complex environments with a large number of states and actions, which can lead to scalability issues.

4 Stability and Convergence

Ensuring the Q-learning algorithm converges to an optimal solution and maintains stable performance.

The Power of Q-Value and Q-Learning



1

Informed Decisions

The Q-value, as the key metric in reinforcement learning, empowers agents to make informed and optimal decisions that maximize long-term rewards.

2

Adaptability

The Q-learning algorithm allows agents to continuously update their knowledge and adapt to changing environments, making it a powerful tool for real-world applications.

3

Transformative Impact

The practical applications of reinforcement learning and Q-learning span diverse industries, from robotics and autonomous vehicles to gaming and healthcare, unlocking new frontiers of innovation.